Reversible Reations

Reversible reaction: A reaction that can be proceed in either direction. The reactants react to form products, and the products react to reform reactants.

→ It's exothermic in one direction, and endothermic in the opposite direction.

To test for water, two reversible reactions can be done:

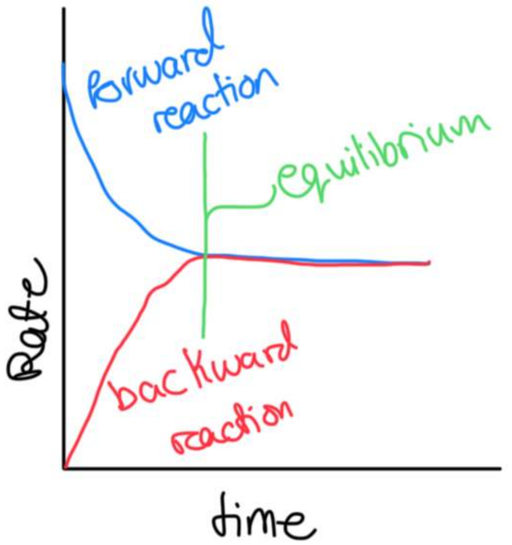
1- Anhydrous copper(II) sulfate is used to test for the presence of water as it changes from white to blue.

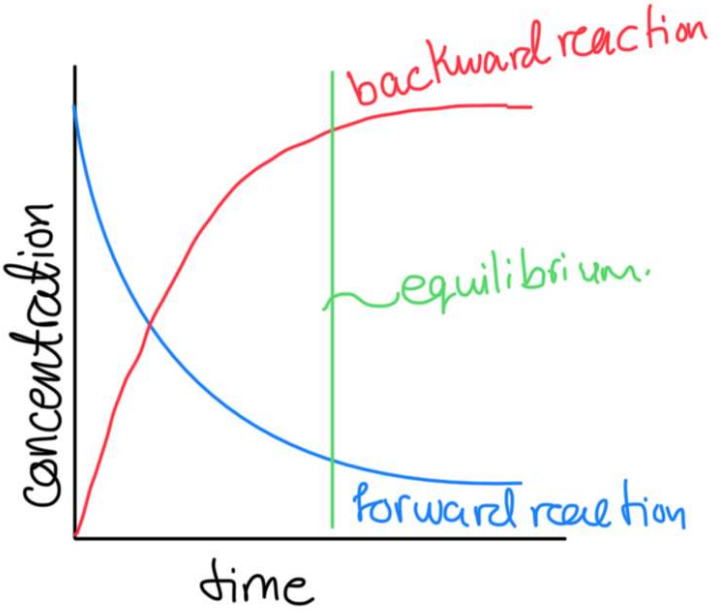
$$\underset{\text{White (s)}}{CuSO_4} + \underset{(l)}{5H_2O} \rightleftharpoons \underset{\text{Blue}}{CuSO_4 \cdot 5H_2O_{(s)}}$$

2- Anhydrous cobalt (II) chloride is used to test for the presence of water as it turns from blue to pink

$$\underset{\text{Blue}}{CoCl_{2(s)}} + \underset{(l)}{6H_2O} \rightleftharpoons \underset{\text{Pink}}{CoCl_2 \cdot 6H_2O_{(s)}}$$

Dynamic equilibrium: When the forward and backward reactions of a reversible reaction in closed system take place at the same rate, so there is no overall change. The concentration of reactants and products do not change





Rate of forward rxn = Rate of backward rxn

Concentration of forward rxn = Concentration by backward rxn.

Conditions for dynamic equilibrium:

1- Closed system: Allows nothing to enter or leave.

2- Should be a reversible

Characteristics of equilibrium:

1- It is dynamic, the reaction has not stopped, the forward and the backward reactions are still taking place at the same rate.

2- The concentration of reactants and products remain constant.

Factors that affect the position of the equilibrium.

1- Changing the concentration: Increasing the concentration of one substance makes the equilibrium shift in the direction that produces less of a substance

* Increasing the concentration of the reactants shifts the equilibrium to the right; favors the product side, increasing the yield. decreasing the concentration of reactants causes the equilibrium to shift to the left, decreasing the yield.

* Increasing the concentration of the products shifts the equilibrium to the left side; favors the reactant side, producing less yield. decreasing the concentration of the products of causes the equilibrium to shift to the right, producing more yield.

2- Changing the temperature of the system: Increasing the temperature of the system makes the reaction move in the endothermic direction.

* Increasing the temperature will shift the equilibrium to oppose this, and move in the endothermic direction to reduce the temp. by absorbing heat.

* Decreasing the temperature will shift the equilibrium to oppose this, and move in the exothermic direction to increase the temperature by giving out heat.

Temperature	Yield	Rate of forward	Rate of backward
Increase	Decrease/ increase	Increase	Increase
Decrease	Increase/ Decrease	Decrease	Decrease

3- Changing the pressure (only in gases)

* Increasing the pressure will shift the equilibrium towards the side with fewer gas molecules.
* Decreasing the pressure the equilibrium shifts towards the side with higher gas molecules.
* If the number of gases on both sides of the equilibrium is equal, increasing or decreasing the pressure has no effect on the equilibrium position.

→ Using high pressure will be dangerous for employees and cost highly due to energy, so companies use moderate pressures.

* Adding a catalyst has no effect on the position of the equilibrium, it will only increase the rate of the reactions.

Haber process: Making ammonia in industry

$$N_{2\,(g)} + 3H_{2\,(g)} \underset{\text{endothermic}}{\overset{\text{exothermic}}{\rightleftharpoons}} 2NH_{3\,(g)}$$

Reactants needed to produce are nitrogen and hydrogen.

To obtain nitrogen: Usually obtained from the atmosphere by fractional distillation of liquid air

To obtain hydrogen: ① The reaction between methane & steam

Methane + Steam $\rightleftharpoons$ Hydrogen + carbon monoxide

$$CH_{4\,(g)} + H_2O_{(g)} \rightleftharpoons 3H_{2\,(g)} + CO_{(g)}$$

Conditions: * Heat * Pressure * Catalyst: Nickel

② Cracking hydrocarbons from petroleum: Alkane: HC

③ Electrolysis of brine, concentrated NaCl solution, where hydrogen is produced at the cathode.

$$2H^+ + 2e^- \longrightarrow H_2$$

Haber Process

Conditions for making ammonia:

1- Temperature: 450°C, (moderate), could be anything between 350°C - 500°C

2- Catalyst: Iron

3- Pressure: 200 atm = 20,000 Kpa

* Under these conditions, the gas mixture leaving the vessel contains only about 15% ammonia, which is then removed by cooling and condensing it is a liquid.
* The hydrogen and nitrogen are recirculated into the reaction vessel to react together once more to produce further quantities.

→ To create more ammonia, higher pressure & lower temperature should be used. But only 200 atm and 450 °C. Because it needs to be ensured that ammonia is made economically

* A pressure of 400 atm needs very powerfull pumps, and very strong and sturdy pipes and tanks. It also needs a lot of electricity, which is expensive and dangerous. So, 200 atm pressure is used because it is safer and saves money.

* Using very low temperatures will slow the rate of reaction, and ammonia is not produced quickly enough for the process to be economical. So, the optimum temperature used is 450°C

* Iron, the catalyst used, speeds up the reaction, but does not change the yield.

→ Removing ammonia will decrease the concentration of ammonia So the equilibrium shifts to the right to make more ammonia

Steps in the contact process:

1- Sulfur dioxide is produced by the reaction of sulfur with air

$$S_{(s)} + O_{2(g)} \longrightarrow SO_{2(g)}$$

2- The gases are heated at a temperature of 450°C, and fed to the reaction vessel, where they are passed to the catalyst vanadium (V) oxide: V_2O_5. It catalyses the reaction between sulfur dioxide and oxygen to produce sulfur trioxide.
sulfur(VI) oxide

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

Conditions for this reaction:

1- Temperature = 450°C

2- Pressure = 1-2 atm / 100-200 Kpa

3- Catalyst : Vanadium (V) oxide. V_2O_5

* 96% yield of SO_3
* Exothermic reaction
* The heat produced by this reaction is used to heat the upcoming reactions, so saves money

3- Sulfur trioxide is dissolved in concentrated sulfuric acid (98%), which gives a substance called oleum.

Sulfur trioxide is not dissolved in water because it reacts very violently, and produces a thick mist of acid which is hard to condense.

$$\text{Sulfuric acid} + \text{Sulfur trioxide} \longrightarrow \text{Oleum}$$

$$H_2SO_{4(aq)} + SO_{3(g)} \longrightarrow H_2S_2O_{7(l)}$$

4- Oleum formed is then added to the correct amount of water to produce sulfuric acid of required concentration.

$$H_2S_2O_{7(l)} + H_2O_{(l)} \longrightarrow 2H_2SO_{4(l)}$$

Properties of ammonia: NH_3:

* Colorless gas with a strong chocking smell
* less dense than air, (collected by upward delivery)
* Reacts with hydrogen chloride to form a white smoke, which is made of tiny particles of solid ammonium chloride. This reaction is used to test for the presence of ammonia.
* Very soluble in water. $NH_{3(g)} + H_2O_{(l)} \longrightarrow NH_4OH$
* Solution in water is alkali, turns red litmus paper blue.
* Reacts with acids to form salts.

Uses of ammonia:

* Making fertilizers
* Making nitric acid
* Making nylon
* Making explosives.

$$2HN_{3\,(aq)} + H_2SO_{4\,(aq)} \longrightarrow (NH_4)_2SO_{4\,(aq)}$$

$$2NH_4OH_{(aq)} + H_2SO_{4(aq)} \longrightarrow (NH_4)_2SO_4 + 2H_2O_{(l)}$$

To test for ammonia:

* Physical test: Turns red damp litmus paper blue
* Chemical test: Reacts with hydrogen chloride gas to form white smoke of ammonium chloride.

Plants need nitrogen, potassium, and phosphorus:

* Nitrogen for making chlorophyll and proteins.
* Potassium produce proteins and resist diseases.
* Phosphorus helps roots to grow, and crops to ripen

Types of fertilizers:

1- Manure: A natural fertilizer

2- Synthetic fertilizers:

* Nitrogenous fertilizers: Where ammonia reacts with something to make synthetic fertilizers.

1- Ammonia reacts with nitric acid to give ammonium nitrate

$$NH_{3(aq)} + HNO_{3(aq)} \longrightarrow NH_4NO_{3(aq)}$$

2- Ammonia reacts with sulfuric acid to give ammonium sulfate.

$$2NH_{3(aq)} + H_2SO_{4(aq)} \longrightarrow (NH_4)_2SO_{4(aq)}$$

3- Ammonia reacts with phosphoric acid

$$NH_{3(aq)} + H_3PO_{4(aq)} \longrightarrow (NH_4)_3PO_{4(aq)}$$

* NPK fertilizers: They contain ammonium nitrate (NH_4NO_3), ammonium phosphate ($(NH_4)_3PO_4$), and potassium chloride (KCl)

Problems with fertilizers:

* If two much fertilizer is applied to the land, rain washes the fertilizer off the land and into rivers and streams.
* May cause aquatic life can survive.

Contact process: To produce sulfuric acid

Sulfur dioxide is a gas that forms when sulfur burns in air. Sulfur dioxide is obteined when sulfide ores, such as lead and zinc ores, are roasted in air

$$2ZnS_{(s)} + 3O_{2(g)} \longrightarrow 2ZnO_{(s)} + 2SO_{2(g)}$$

Properties of sulfur dioxide:

1- It's a colorless gas, heavier than air (so collected by downward delivery) with a strong, chocking smell.

2- An acidic oxide, that dissolves in water to form sulfurous acid, H_2SO_3 $\hookrightarrow$ By sulfite (SO_3^{2-})

$$H_2O_{(l)} + SO_{2(g)} \longrightarrow H_2SO_{3(aq)}$$

3- It acts as bleach when it is damp or in solution, because it removes the color from colored compounds by reducing them.

4- Kills bacteria.

Uses of sulfur dioxide:

* Manufacture of sulfuric acid
* To bleach wool, silk, and wood pulp for making paper.

Main sources of sulfur dioxide:

Coals and petroleum that burn in power station, and factory furnaces, being oxidized to sulfur dioxide

Harms of sulfur dioxide:

- Causes breathing problems
- Dissolves in rain to cause acid rain. This attacks buildings and metal structures and can kill fish and plant

To test for sulfur dioxide:

- Turns damp blue litmus paper.
- Turns acidified potassium manganate (VII) solution from purple to colorless

Contact process

Raw materials for making sulfuric acid:

- Sulfur / sulfur dioxide
- Air
- Water
- Concentrated H_2SO_4

* The reaction to produce SO_3 is exothermic, which is favored by low temperature. However, the reaction would be too slow to be economic, so the optimum temperature used is 450°C, with vanadium (V) oxide as a catalyst to increase the rate on both reactions, so equilibrium is reached faster.

* Increasing pressure favors the side with fewer gas molecules, so the equilibrium shifts to the right, increasing the yield of SO_3. But the process is already run at 1-2 atm, and there is no need to increase the pressure because the yield is already high (96%)

Uses of sulfuric acid:

* Making detergents
* Powerful oxidizing agent.
* As a catalyst
* As a dehydrating agent: It is a powerful dehydrating agent, it will take the water from a variety of substances.

$$\text{Sucrose} \xrightarrow{\text{conc } H_2SO_4} \text{Carbon} + \text{Water}$$

$$\underset{(s)}{C_{12}H_{22}O_{11}} \longrightarrow 12C_{(s)} + 11H_2O_{(l)}$$

$$\text{Hydrated copper sulfate} \xrightarrow{H_2SO_4} \text{Anhydrous copper(II) sulfate} + \text{Water}$$

$$\underset{(s)}{CuSO_4.5H_2O} \longrightarrow \underset{(s)}{CuSO_4} + 5H_2O_{(l)}$$

$$\text{Blue} \longrightarrow \text{Colorless}$$

* The reaction is two highly exothermic

* Ammonia cannot be dried by concentrated sulfuric acid because it is an alkali, and it will react with the acid.

*Dilute sulfuric acid is made by adding the concentrated H_2SO_4 to water, with cooling. Never add the water to the acid, because it would produce so much heat, and the acid could splash out and burn you.

Uses of dilute sulfuric acid:

1- Making fertilizers.

2- Making paints, dyes and fibres.

3- A common labratory reagent.

To test for sulfate:

Add a few drops of dilute hydrochloric acid, to neutralize and remove and carbonate ions that would produce false white precipitate results. Then add few drops of barium chloride. A whit precipitate of barium carbonate is produced.

Barium ions + Sulfate ions ⟶ Barium sulfate

$$Ba^{2+}_{(aq)} + SO_4{}^{2-}_{(aq)} \longrightarrow BaSO_{4(s)}$$